

Q1. Transfer Functions

- (a) Calculate the transfer function relating V_o to V_i for the circuit in Figure 1 and plot the pole-zero diagram.

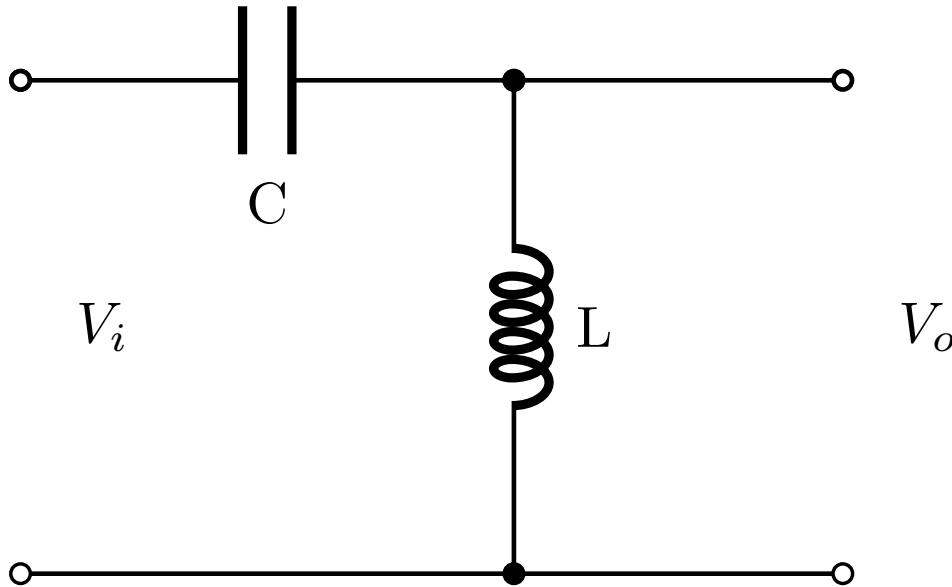


Figure 1:

- (b) Calculate the transfer function relating V_o to V_i for the circuit in Figure 2 and plot the pole-zero diagram.

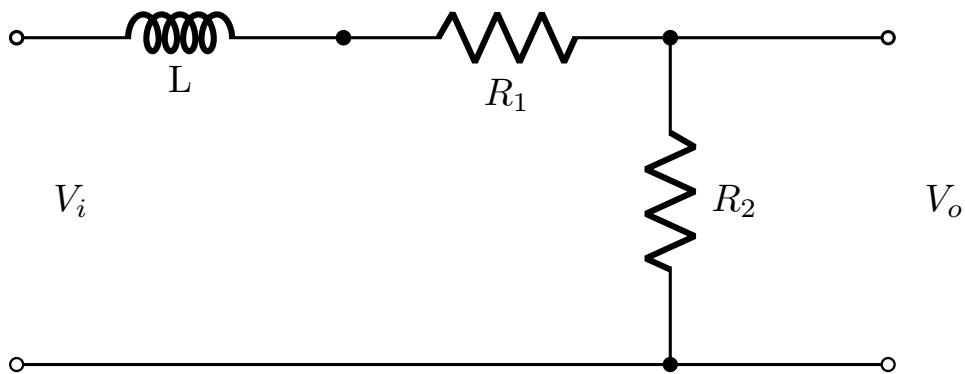


Figure 2:

Q2. Single Pole Low-Pass Filter

Consider the signal:

$$x(t) = \cos(2\pi t) + \cos(1000\pi t) \quad (1)$$

input to a filter with an s-domain transfer function

$$H(s) = \frac{0.5}{s + 0.5}. \quad (2)$$

- (a) Determine the magnitude of the frequency response of the filter $|H(f)|$.
- (b) Determine the 3 dB cut-off frequency.
- (c) Sketch the magnitude of the frequency response of the filter.
- (d) Sketch the Fourier transform of the signal, $X(f)$.
- (e) Determine how much the higher frequency component is attenuated compared to the lower frequency component.
- (f) Sketch the approximate Fourier transform of the output of the filter $Y(f)$.
- (g) Sketch the approximate output of the filter $y(t)$.
- (h) Consider a similar transfer function:

$$H(s) = \frac{1}{s + 0.5}, \quad (3)$$

and explain why it cannot be achieved with a passive filter.

Q3. RC Anti-alias filter

A first order RC anti-aliasing filter is required with a (3 dB) cut-off frequency of 20 kHz.

- (a) Sketch the circuit.
- (b) Determine the s-domain transfer function $H(s)$.
- (c) Determine the magnitude of the Fourier domain frequency response $|H(f)|$.
- (d) What resistor value is required to achieve the desired cut-off frequency given a 1 nF capacitor.
- (e) Determine how much attenuation the filter provides at 2 MHz.
- (f) If more attenuation were required at 2 MHz, what would you do?

Q4. High-Pass Filter

Consider a high-pass filter comprised of a $1\text{ k}\Omega$ resistor and a 1 nF capacitor.

- (a) Sketch the circuit.
- (b) Determine the s-domain transfer function $H(s)$.
- (c) Determine the cut-off frequency.
- (d) Sketch the magnitude of the frequency response of the circuit $|H(f)|$.
- (e) For frequencies well below the cut-off frequency, determine the phase shift of the filter.